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CRITICAL PERSPECTIVES

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Pharmacotherapies for Gaucher disease: what's new on the horizon?

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ABSTRACT

Introduction: Gaucher disease (GD) is an autosomal recessive lysosomal storage disorder that results in the accumulation of glucosylceramide (GluCer) within macrophages. Current treatments for GD include enzyme replacement therapy (ERT) or substrate reduction therapy (SRT) to reduce the total amount of GluCer and improve organ function.

Areas covered: We review the two main FDA approved therapies for GD, ERT and SRT, including advances in ERT/SRT delivery and new clinical trials using Pubmed and trial databases. New and emerging therapies for GD are discussed in depth, including pre-clinical and clinical studies of small molecule drugs, gene therapy, and nanotechnology. An overview of current clinical trials for GD is provided.

Expert opinion: Current therapies have vastly improved the health and quality of life for patients. Future efforts in therapeutic development should focus primarily on brain penetrant therapies that will ameliorate neuronopathic GD as well as *GBA1*-related Parkinson disease. These therapies should be assessed using specific biomarkers that are well validated through advances in translational research.

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1. Introduction

Gaucher disease (GD) is a rare autosomal recessive lysosomal storage disorder (LSD) resulting from pathological variants in the gene *GBA1*, which encodes the lysosomal enzyme glucocerebrosidase (GCase). GCase deficiency causes an accumulation of substrates glucosylceramide (GluCer) and glucosylsphingosine (GluSph) within reticuloendothelial cells. GluSph, also known as Lyso-Gb1, has been correlated with disease severity and is a key biomarker for GD [1,2]. The primary clinical manifestations of GD include hepatosplenomegaly, thrombocytopenia, anemia, and skeletal involvement [3]. Gaucher disease is categorized into three main types. Type 1 GD (GD1) is characterized by the absence of primary neurological involvement, Type 2 GD (GD2) is characterized by accumulation of substrate during fetal development [4], leading to severe neurodegeneration and typically resulting in death in infancy [5]. Type 3 GD (GD3) includes both neurological and visceral manifestations and has a high phenotypic variability, especially regarding neurological features [6].

The wide range of pathogenic *GBA1* variants encountered in patients with GD accounts for some, but not all the observed phenotypic variation [7]. Most *GBA1* variants are missense [5], which lead to misfolding of GCase in the endoplasmic reticulum (ER). Other mutant forms of GCase are not properly trafficked to the lysosome [8]. Some patients carry null *GBA1* alleles, where no functional GCase is produced [9].

There are currently five FDA-approved therapies for Gaucher disease, which fall into two broad categories: enzyme replacement therapy (ERT) and substrate reduction therapy (SRT). ERT involves intravenous (IV) delivery of a recombinant mannose-terminated GCase which serves as a substitute for endogenous enzyme. SRT, an orally administered small molecule drug, lowers the amount of GluCer within lysosomes by inhibiting glucosylceramide synthase, the enzyme that catalyzes the conversion of ceramide to GluCer [10]. While both types of therapy are effective against visceral and hematological complications of GD, they are ineffective against neuronopathic GD (nGD) as they do not penetrate the blood–brain barrier (BBB) [11].

In addition to the approved therapies for GD, the investigational landscape is broad, including many novel therapies currently in clinical trials. These include derivations of existing treatments, such as Venglustat, a new central nervous system (CNS)-penetrant SRT [12]. In addition, efforts are underway to expand the scope of SRT and ERT, including in pediatric and fetal populations [13–15]. Several new strategies for the treatment of GD have shown promise in pre-clinical studies. Small molecule therapies, including BBB-penetrant compounds, have been identified through screening, leading to the identification of potential candidate drugs [15]. Gene therapy has also been extensively evaluated in animal models of GD, and several active clinical trials have begun [16].

GBA1 variants have also provided key insights into the role of lysosomal dysfunction in Parkinson disease (PD), a common neurodegenerative disorder. Individuals carrying variants in

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Article highlights

- *GBA1* variants cause Gaucher disease (GD) and are the most common genetic risk factor for Parkinson disease.
- Current therapies for GD include substrate reduction and enzyme replacement therapies. These have been highly effective in patients with type 1 GD but are ineffective for neuronopathic GD.
- Multiple new strategies are being pursued, including small molecule drugs and gene therapy trials.
- Future CNS-penetrant therapeutics may have utility for both neuronopathic GD and *GBA1*-associated PD.

GBA1, including heterozygous carriers, have an elevated risk for developing PD [17]. The *GBA1*-PD phenotype is now widely recognized, with *GBA1 v*ariants considered the most common known genetic risk factor for PD, although the penetrance remains low. Furthermore, this association has sparked emerging therapeutic strategies targeting GCase pathways [18].

This review covers both new and emerging therapies for Gaucher disease, with special focus on pre-clinical and early-stage clinical therapeutics (Figure 1). Relevant literature was sourced using Pubmed and clinical trial databases from November 2025 to April 2026, with priority given to comprehensive reviews, original research, and clinical trial results. Only articles published in English were included.

2. Enzyme replacement therapy

Treatments for GD prior to ERT addressed the symptoms of GCase deficiency, such as splenomegaly and bone manifestations, rather than the deficiency itself. In 1965, researchers provided evidence of the central role of GCase deficiency in GD, suggesting enzyme supplementation as a therapeutic avenue [19]. Using purified human placental GCase, investigators demonstrated that exogenous GCase could decrease glucocerebroside accumulation in patients with GD [20,21]. Adding mannose residues reportedly improved uptake by macrophages, leading to the development of alglucerase in 1991, the first ERT approved for an LSD [22] (Figure 2).

Limitations in the availability of human placentas and fear of the potential transmission of pathogens highlighted the need for a feasible alternative. Subsequent sequencing and cloning of the *GBA1* gene allowed for recombinant DNA to be produced from Chinese Hamster Ovary (CHO) cells [23–25]. Imiglucerase, recombinant human GCase with a single amino acid difference, was approved in 1994 for GD1, replacing alglucerase (Figure 2). Velaglucerase alfa, developed in human fibroblast cell lines with the native human enzyme sequence, was approved for GD1 in 2010 [26] (Figure 2). Taliglucerase alfa, a recombinant plant-derived GCase produced in carrot cells with the same Arg495His amino acid difference as imiglucerase, was approved in 2011 [27,28] (Figure 2). Studies evaluating dosing based on disease severity are limited, but one study has shown that extending the treatment interval to 3–4 weeks was non-inferior to the standard biweekly infusions [29]. One potential concern with protein-based-therapeutics like ERT is the development of antibodies to the drugs. While antibodies have been detected in 2–15% of treated individuals, treatment outcomes are rarely affected, and neutralizing antibodies are very rare [30].

ERT is effective at relieving systemic manifestations, dramatically reducing organomegaly and improving hematological parameters, but its inability to cross the BBB limits its effectiveness against neurological involvement present in GD2 and GD3 [31]. *In-utero* delivery of ERT, which would cross the

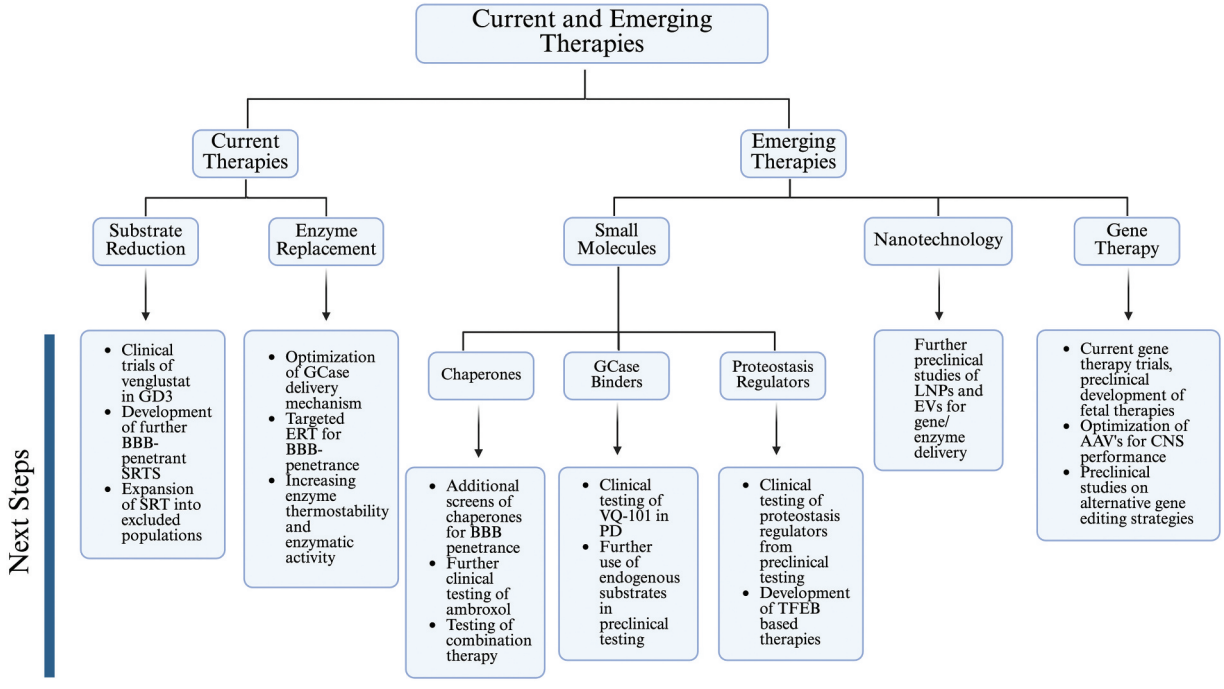


Figure 1. Schematic illustrating the different therapeutic strategies available and being developed for the treatment of Gaucher disease and *GBA1*-associated Parkinson disease.

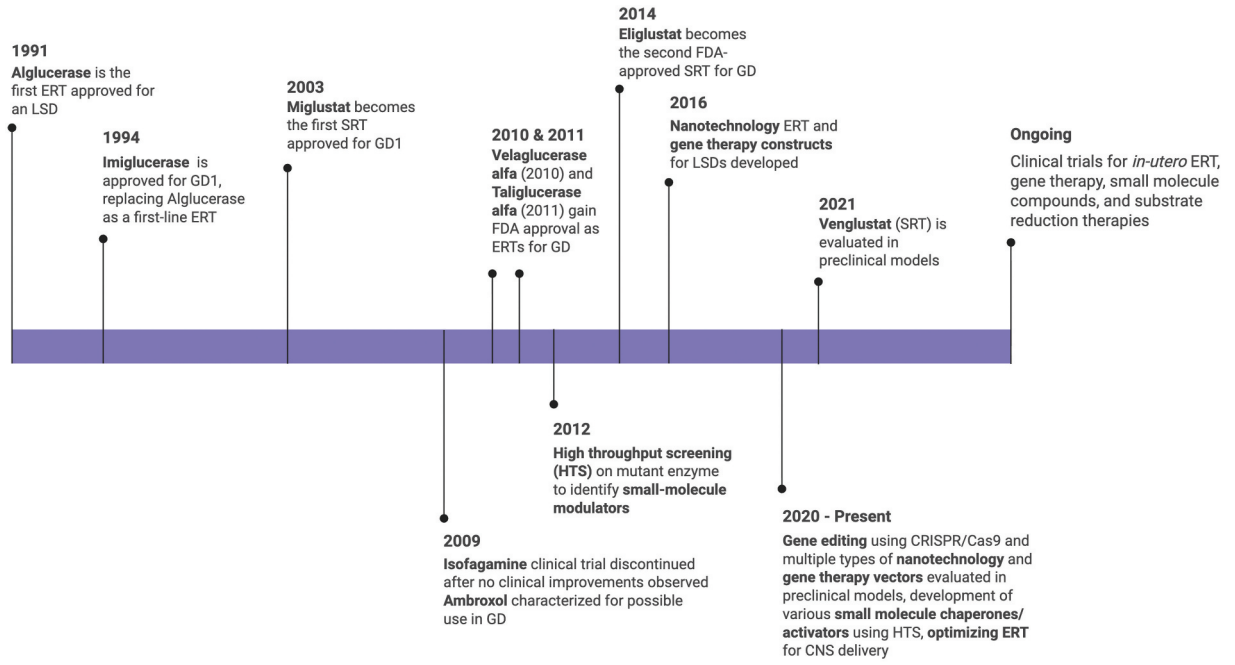


Figure 2. Timeline summarizing the pace of drug development for Gaucher disease over the past four decades.

developing BBB to support normal neuronal development and alleviate visceral pathology, has been proposed [14,32]. The phase I clinical trial, PEARL, is currently recruiting patients to determine the safety and feasibility of *in-utero* ERT in fetuses with LSDs [32] (Table 1). However, a means to continue to target the enzyme to the CNS postnatally remains necessary.

Efforts to optimize GCase activity and delivery are underway. A more thermostable form of GCase with higher enzymatic activity compared to human GCase has been designed using the PROSS stability-design method [33]. Stereotaxic injection of the protein via AAV vector into GD3 mice improved motor performance and life span and decreased neuroinflammation [34]. The transferrin receptor (TfR) transports the iron-binding protein transferrin across the BBB and is being exploited for CNS ERT delivery [35]. Fusion of TfR to recombinant human and murine GCase mediated transport across the BBB and improved GCase delivery to lysosomes. One brain shuttle construct developed in Japan has been approved for Mucopolysaccharidosis Type 1 [36], suggesting its potential could be harnessed for GD.

The high cost, inconvenience of regular IV infusions, and lifelong need for ERT have highlighted the need to develop alternative therapies. Many patients have switched to home ERT infusions to alleviate burden of frequent infusions [37]. Some patients, especially children, have had ports placed to ease venous access, but these are not without long-term risks. Oral medications may be advantageous for these patients.

3. Substrate reduction therapy

The FDA has approved two oral SRTs: miglustat in 2003, and eliglustat in 2014 [11]. Unlike ERT, they are only approved for adults [13]. Their primary mechanism of action is to reduce substrate burden to compensate for decreased catabolic capacity due to GCase deficiency. This is accomplished by inhibiting glucosylceramide synthase, an enzyme that catalyzes the synthesis of GluCer [38].

Early clinical trials of miglustat showed reduction of organomegaly and improvements in hematological symptoms, although side effects included diarrhea and tremor [39]. Hoping that SRT might provide a viable therapy for nGD [40], Schiffman and colleagues conducted the first trial of miglustat in GD3, choosing saccadic eye movement velocity as a measure of GD-related neurological impairment [41]. No improvement in saccadic velocity or other neurological symptoms was observed, suggesting that despite potential BBB-permeability, miglustat is either ineffective in treating nGD, or the outcome parameter, saccadic eye movement velocity, was not appropriate [42].

Eliglustat functions through a similar mechanism to miglustat but is metabolized by the CYP2D6 cytochrome P450 enzyme and therefore requires assessment of the patient's CYP2D6 metabolizer status prior to use. Only adults who are extensive, intermediate, or poor metabolizers are eligible for eliglustat [43]. Multiple long-term clinical trials demonstrated comparable efficacy between eliglustat and imiglucerase both in patients who switched from ERT prior to eliglustat, and treatment-naïve patients [43–45]. As a result, eliglustat has been approved as a first-line therapy for GD1, while miglustat is currently a second-line therapy for patients unable to tolerate ERT.

Eliglustat also acts as an inhibitor of cardiac ion channels, including Nav1.5 and Cav1.2 [46], raising controversy regarding its relative cardiac safety profile [47,48]. A recent trial dosed eliglustat based on CYP2D6 status in patients with GD1 and concurrent cardiac comorbidities, demonstrating high safety and efficacy and supporting pharmacogenomic approaches to guide safe dosing in this population [46]. Other trials are underway to expand the scope of eligible

Table 1. Current clinical trials for Gaucher disease and *GBA1*-associated Parkinson disease.

Intervention/Treatment name (Sponsor name)	Intervention type	Patient criteria	Study phase (Study name)	Recruitment status	Identifier	Location
Aldurazyme, (University of California, San Francisco)	*In-utero* enzyme replacement therapy	Pregnant women carrying a live male or female fetus with an LSD at 18 0/7 weeks to 34 6/ 7 weeks gestation.	Phase 1 (PEARL)	Recruiting	NCT04532047	San Franscisco, CA, USA
BIA 28–6156 (Bial R&D Investments, S.A)	Small molecule chaperone	PD with *GBA1* variant, ages 35–80 years	Phase 2 (ACTIVATE)	Active, not recruiting	NCT05819359	95 locations across U.S.A., Canada, France, Germany, Italy, Netherlands, Poland, Portugal, Spain, Sweden, UK
YH35995 (Yuhan Corporation)	Substrate reduction therapy	Healthy males, ages 19–45 years	Phase 1	Recruiting	NCT06517914	Seongnam, South Korea
Venglustat (Sanofi)	Substrate reduction therapy	GD3 adults ≥18 years of age or children ages 12–18, on ERT for at least 3 years	Phase 3 (LEAP2MONO)	Active, not recruiting	NCT05222906	22 locations across U.S.A., Canada, Argentina, China, France, Germany, Hungary, Italy, Japan, Turkey, UK
PR001/LY3884961 (Prevail Therapeutics)	Gene therapy – AAV9	GD1, ages 18 years and older, on ERT or SRT previously	Phase 1/2 (PROCEED)	Recruiting	NCT05487599	Chicago, IL, U.S.A.; Durham, NC, U.S.A.; Fairfax, VA, U.S.A.; New South Wales, Australia; Rio Grande do Sul, Brazil; Höchheim, Germany; Zaragoza, Spain; London, UK
		GBA1-PD, ages 35–80 years	Phase 1/2 (PROPEL)	Recruiting	NCT04127578	15 locations in U.S.A. and Israel
		GD2 infants, ages 0–24 months	Phase 1/2 (PROVIDE)	Active, not recruiting	NCT04411654	Oakland, CA, U.S.A.; Minneapolis, MN, U.S.A.; Pittsburgh, PA, U.S.A.; Fairfax, VA, U.S.A.; Manchester, UK
VGN-R08b (Xinhua Hospital, Shanghai Jiao Tong University School of Medicine/Shanghai Vitalgen BioPharma Co., Ltd.)	Gene therapy – AAV9	GD2 infants, ages 0–24 months	Early phase 1	Recruiting	NCT06272149	Shanghai, China
		GBA1-PD, ages 30–70	Phase 1/2	Not yet recruiting	NCT07414290	N/A
LY-M001 (Lingyi Biotech)	Gene therapy – AAV8	GD1, ages 6–18 years	Phase 1/2	Active, not recruiting	NCT06528080	Shanghai, China
		GD1, ages 18–60 years	Phase 1/2	Recruiting	NCT06818838	Guangzhou, China; Shanxi, China; Tianjin, China
FLT201 (Spur Therapeutics)	Gene therapy – AAVS3	GD1, ages 18 years and older	Phase 1/2 (GALILEO-2)	Active, not recruiting	NCT06545136	Fairfax, VA, U.S.A.; Porto Alegre, Brazil; Zaragoza, Spain; Salford, UK
		GD1, ages 17 years and older, on ERT or SRT previously	Phase 3 (GALILEO-3)	Not yet recruiting	NCT07223944	N/A

patients for eliglustat, including in pediatric populations. The ELIKIDS trial (NCT03485677) aims to test the efficacy of eliglustat with and without imiglucerase in children with GD1 and GD3, but so far, no long-term data on treatment in children exists. Despite these recent advances, eliglustat is still contraindicated for individuals on a number of drugs, including any strong inhibitors of CYP2D6 [49]. Additionally, eliglustat is contraindicated during pregnancy, based on early animal studies suggesting the potential for developmental abnormalities in exposed fetuses [50].

Efforts to develop BBB-permeable SRT molecules capable of modifying disease progression in nGD led to venglustat, shown in preclinical studies to be BBB-permeable and effective in reducing brain glycosphingolipid levels in a mouse model of GD [51]. Additionally, venglustat was considered for *GBA1*-PD, supported by murine studies where it reduced the α-synuclein burden and improved cognitive function [51]. Despite this, in the MOVES-PD trial, venglustat had no beneficial effect on *GBA1*-PD [52]. There are currently several other trials focusing on venglustat in GD3, including the LEAP trial with imiglucerase (NCT02843035), and the LEAP2MONO trial compared to imiglucerase (NCT05222906; Table 1). Preliminary results from the LEAP trial suggest a tolerable safety profile and improvements in qualitative patient-reported outcomes [53], although a Phase 2 study reported decreased cognitive performance despite treatment, based on neuropsychological testing [54]. Further studies are needed to determine the effectiveness of venglustat in GD3.

4. Small molecule therapies

Developing an orally available small molecule therapy that targets GCase function and can cross the BBB would be desirable for neuronopathic for GD and potentially for *GBA1*-associated PD. Several strategies are being actively pursued.

4.1. Proteostasis regulators

Proteostasis refers to the controlled concentration, conformation, binding interactions, and subcellular location of proteins through an extensive network, ensuring a stable and functional proteome [55]. The balance between protein folding and degradation is tightly regulated, and loss of proteostatic control can induce stress pathways. Most pathogenic variants

in *GBA1* are single amino acid substitutions, which impair enzyme folding and destabilize the protein's native folding, leading to endoplasmic reticulum associated degradation (ERAD) of the misfolded enzyme [56]. Mutant GCase has been shown to increase mRNA and protein levels of unfolded protein response (UPR), indicating dysregulation of proteostasis [57].

Proteostasis regulators (PRs) are indirect modifiers of GCase which aim to remodel the proteostasis network by targeting posttranslational processes to promote native folding and lysosomal activity of mutant GCase enzymes, either through an enhancement of folding, a reduction in degradation, or a modification of transcription [58].

4.1.1. Enhancing ER-associated folding (ERAF) and decreasing ERAD

Inducing molecular chaperones to promote folding of mutant GCase is one proteostatic approach. Celastrol has been shown to increase GCase activity by stabilization of the mutant form of the enzyme and through inhibition of ERAD [59]. Heat-shock protein amplifier Arimoclomol was shown to induce the heat shock response and promote the folding, maturation, activity and lysosomal trafficking of mutant GCase [60]. Arimoclomol, which crosses the BBB, is FDA-approved for the LSD Niemann-Pick C.

Intracellular free cholesterol induces ER stress, depletes Ca^{2+} stores, triggers the UPR and apoptosis and can moderate the ERAD of GCase, with higher levels found in some patients with GD [61,62]. Mevastatin, an HMG-CoA reductase inhibitor, decreased ERAD and increased the stabilization, maturation, activity and lysosomal translocation of mutant GCase, indicating a possible therapeutic route for improving proteostasis in GD patients [62].

GluCer accumulation can cause excessive Ca^{2+} efflux through ryanodine receptors (RyR) in the ER and impair Ca^{2+} homeostasis, which can affect the folding of unstable GCase variants [63]. Reestablishing the Ca^{2+} gradient and lowering cytosolic Ca^{2+} improved the native folding of GCase and increased GCase activity [63]. Decreasing Ca^{2+} efflux through RyR and increasing ER Ca^{2+} levels also enhanced GCase homeostasis [64].

Drugs inhibiting L-type Ca^{2+} channels have been explored. Verapamil and Diltiazem partially restored GCase homeostasis in N370S (c.1226A > G, p.Asn409Ser) and L444P (c.1448T > C, p.Leu483Pro) GD fibroblasts, and Diltiazem also enhanced ER folding [65]. Lacidipine inhibited ER efflux and prevented apoptosis induction, while also enhancing folding, trafficking, and activity of mutant GCase [66]. Eeyarestatin I (EerI), an inhibitor of retrotranslocation of misfolded substrates to cytoplasm, blocks ERAD in GD cells and prolongs ER retention of mutated GCase while inducing the UPR and apoptosis [56]. Application of Lacidipine in combination with EerI protected GD cells by activating the anti-apoptotic signaling cascade, enhancing ER folding and decreasing ERAD of GCase variants [56].

4.1.2. Regulation of transcription and translation

Transcription factor EB (TFEB), which regulates the autophagy-lysosomal pathway, is negatively regulated by the mammalian target of rapamycin complex 1 (mTORC1), which regulates cell growth and metabolism [67]. Dysregulation of the autophagy-lysosomal pathway is implicated in neuronopathic forms of GD. Increased phosphorylation of TFEB by mTORC1 been shown to mediate lysosomal dysfunction in GD cells [67]. TFEB activity was decreased in PD patient derived iPSC dopaminergic neurons, and inhibition of mTOR, a component of mTORC1, was able to restore TFEB activity and decrease ER stress and α-synuclein accumulation [68].

4.2. Direct GCase binders

As PRs are not direct GCase binders, they may not be optimal for restoring GCase activity, folding, or trafficking [69]. Direct GCase binders can improve the activity or folding of the mutant enzyme, providing a more effective therapeutic approach.

4.2.1. Allosteric activators

Allosteric activators directly bind to GCase and alter the active site conformation to improve affinity, thereby increasing the intrinsic activity of the enzyme. These activators may be especially useful in cases of GCase haploinsufficiency, for molecules that increase the catalytic efficiency of protein expressed from the remaining wild-type (WT) allele.

An affinity-based fluorescence polarization screen of FDA-approved drugs identified the antipsychotic quetiapine as a GCase binder in iPSC-derived dopaminergic neurons (DANs) from a PD-patient line heterozygous for the 84GG (c.84dupG, p.Leu29Alafs *18) frame-shift mutation [70], a variant which results in the complete loss of mutant GCase. This line also displays reduced WT protein from the single copy of the WT GCase allele. Quetiapine was able to increase GCase levels and activity, decrease intracellular GluCer levels, and decrease α-synuclein levels in these neurons.

Most drug screens use artificial substrates as a read-out for GCase activity, which may not translate to GCase hydrolysis of native substrates. A drug screen using endogenous GluCers as substrates identified two compounds bound to distinct binding sites of GCase, one of which was separate from the active site. Both compounds activated GCase and increased substrate turnover [71]. This assay may be useful to identify future compounds targeting GCase enzymatic activity.

More recently, preliminary results showed the small molecule allosteric activator VQ-101 (Vanqua) was able to increase GCase activity in *GBA1*-PD-derived DANs, and block α-synuclein accumulation [72]. In a phase Ia/Ib trial, VQ-101 showed full CNS penetrance and dose-dependent increases in GCase activity in PD patients with or without *GBA1* variants. Its effect on PD progression is currently being explored.

4.2.2. Pharmacological chaperones (PCs)

Certain mutant forms of GCase may remain enzymatically active but fail to reach the lysosome, limiting the effectiveness of an activator. PCs assist in the proper folding of the protein and stabilize the enzyme to prevent further misfolding, thereby improving trafficking from the ER to Golgi and lysosome [73]. Two types of PCs, inhibitory and non-inhibitory chaperones, are being developed for GD.

4.2.3. Inhibitory chaperones

The first chaperone considered for GCase was the iminosugar N-(n-nonyl)deoxynojirimycin (NN-DNJ), reported to increase GCase activity in N370S mutant fibroblasts [74]. DNJ also inhibits α-glucosidase and ceramide glucosyltransferase, and this lack of target specificity led to the development of isofagomine, which promoted GCase folding and lysosomal trafficking by binding to its active site [75]. However, in a phase II clinical trial, no clinical improvements were observed with isofagomine, and the trial was discontinued in 2009 [76]. Another iminosugar derivative that binds to both GCase and α-synuclein is being explored for *GBA1*-PD [77].

Ambroxol, a drug used to treat airway mucus hypersecretion diseases including cystic fibrosis, asthma, and hyaline membrane disease in newborns, was identified as a potential chaperone of GCase through a targeted screen performed in a library of approved drugs [78]. It acts as a pH-dependent mixed inhibitor of GCase and interacts with both active and non-active site residues of GCase. Ambroxol was found to increase GCase levels and activity while decreasing GluCer storage in GD lymphoblasts and fibroblasts [78]. Some improvements in neurological manifestations in patients with GD3 have been reported [79]. In untreated GD1 patients, ambroxol showed improvement in disease-specific parameters without severe adverse effects [80]. In a clinical trial of 17 PD patients, ambroxol was found to cross the BBB and increase GCase levels in individuals with and without *GBA1* variants [81]. Recent results from a phase II clinical trial indicate that ambroxol was safe, well tolerated, and increased GCase levels in patients with PD-related dementia [82]. However, only eight of the patients carried a *GBA1* variant, and larger trials with more *GBA1* carriers and homozygotes are needed. The AMBITIOUS (NCT05287503) study, which assessed the effect of ambroxol on cognitive and motor function, GCase activity levels, and α-synuclein levels in patients with *GBA1*-PD, was completed in 2024, but results are not currently available.

4.2.4. Non-inhibitory chaperones

With inhibitory chaperones, finding a dose that balances between its chaperoning effect and inhibition of GCase activity is challenging, as the chaperone must first be released from the active site for GCase to hydrolyze its substrates [69]. In contrast, non-inhibitory chaperones assist the mutant enzyme in proper folding and trafficking by binding to an allosteric site.

A high-throughput screen using mutant N370S GCase identified NCGC758 and NCGC607 as small molecule non-inhibitory chaperones, and further studies in GD iPSC-derived macrophages and DANs, respectively, showed their efficacy in chaperoning mutant enzyme and restoring lipid and GCase activity levels [83–85]. A derivative of the NCGC758 molecule (BIA 28–6156, Bial) is currently in a phase II clinical trial for patients with *GBA1*-PD (Table 1).

GT-02287 and GT-02329 (Gain Therapeutics) are two structurally targeted allosteric regulators (STARs) of GCase, developed using the Site-directed Enzyme Enhancement Therapy (SEE-tx) platform [86]. Both were found to allosterically bind to GCase and improve folding, increasing GCase activity and function. In healthy volunteers, GT-02287 was detected in CSF and plasma at projected therapeutic levels and GCase activity increased after administration [87], and it is currently being tested in a phase 1 clinical trial for PD patients with or without *GBA1* variants. GT-02216, another compound developed by Gain Therapeutics, was able to bind directly to GCase and reduce Tau accumulation, indicating a possible mechanism of action in patients with *GBA1*-PD [88].

Nanobodies (Nbs), small, stable fragments derived from camelid heavy-chain antibodies, are being explored for their ability to target GCase and act as both an allosteric chaperone and catalytic activator. Nbs have been shown to bind to a novel allosteric pocket, highlighting a potential avenue for the development of other small molecule therapies [89].

A pipeline of high-throughput cell-based assays was recently developed to identify small molecule stabilizers of misfolded GCase, with the potential to distinguish between chaperoning and activating ability [15]. Several lead non-inhibitory compounds were identified. Future screens can use this pipeline to validate hit compounds for pure chaperone action.

5. Nanotechnology

The use of nanotechnology to encapsulate compounds offers stability, extended circulation and bioavailability, and the possibility for targeting compounds to specific tissues, cells, or organelles [90]. Nanoparticle carriers could enable crossing of the BBB or increase the half-life of recombinant enzyme (thereby decreasing the frequency of infusions) and can be designed to carry other complex molecules and therapeutic proteins. Nanoparticle carriers can be created from many sources, including proteins, synthetic particles, extracellular vesicles, and virus capsids.

Two main types of synthetic nanoparticles are well-suited for brain delivery: polymeric and lipid nanoparticles (LNPs). Biodegradable polymers for nanoparticles have been FDA-approved for human use and prevent drug degradation, offering a higher drug loading capacity and extended controlled release [90]. Encapsulation of velaglucerase alfa in polymeric nanoparticles was shown to increase GCase activity in treated GD mesenchymal stem cells and enzyme release was only observed at the acid pH of lysosomes [91]. Colocalization of these nanoparticles with lysosomes was noted, suggesting that the enzyme reaches the lysosomes, and the encapsulation prevents the loss of enzyme in circulation. LNPs have low cytotoxicity and immunogenicity, can conjugate to targeting moieties, and are typically utilized for rapid intracellular delivery [92]. Nanoparticles that can cross the BBB are of particular interest for neuronopathic GD and *GBA1*-PD. Saposin C (SapC) is a lysosomal glycoprotein important for GCase activity and function, and nanovesicles containing SapC coupled with the phospholipid dioleoylphosphatidylserine (DOPS) are able to cross the BBB and enhance mutant GCase function [93]. These SapC-DOPS nanovesicles can deliver recombinant enzyme (SapC-DOPS-GCase) into the CNS via a phosphatidylserine-mediated pathway and improve neuroinflammation and degeneration. Their lipid bilayer could protect GCase from pH inactivation and clearance peripherally

enhancing GCase uptake into cells. SapC-based vesicles can also fuse with cell membranes, allowing the enzyme to enter organs inaccessible to available ERTs, reducing inflammation in these areas, an important improvement.

LNPs have also been used to encapsulate mRNA, allowing targeted protein production without genomic editing [94]. An encapsulated synthetic mRNA encoding murine GCase (JCXH-301) in LNPs, administered intravenously to wild-type and D427V mice, displayed a longer half-life than Cerezyme, reached multiple tissues, restored active GCase expression, and reduced GluSph levels. Elevated *GBA1* mRNA levels were detected in the brain, suggesting utility of this drug for neuronopathic GD. Generally, LNP-delivered RNA therapies are much cheaper than gene therapy or current ERT, potentially offering a more sustainable option for patients.

Non-synthetic nanoparticles are also being explored as a therapeutic option. Extracellular vesicles (EVs) are natural vesicles carrying active biomolecules, which can be engineered in large quantities and loaded with therapeutic compounds after production. EVs from *GBA1*-overexpressing HEK293T cells increased GCase levels and activity, were taken up by THP-1 macrophages, and delivered functional GCase to SH-SY5Y neuroblastoma cells [95]. Treating GD3-derived iPSC macrophages with these EVs improved GCase activity. EVs containing GCase were incorporated into the lysosomal compartment of GD3-derived iPSC neurons, rescuing defective lysosomal phenotypes and GCase activity, although their ability to cross the BBB has not been tested *in vivo*.

Virus-like nanoparticles (VLPs) are noninfectious nanoparticles made from a viral capsid. Coat proteins from the capsid can assemble into VLPs that encapsulate the protein inside, a strategy harnessed to improve the stability and half-life of GCase [96].

6. Gene therapy

Gene therapy has long been considered as a therapeutic option for many diseases associated with loss-of-function variants. Therefore, LSDs such as GD are ideal targets for gene therapy as they are single-gene disorders with a well-characterized pathophysiology [97]. Furthermore, earlier studies have shown that bone marrow transplantation can reverse the hematologic and visceral manifestations of GD, providing the proof-of-principle that bone marrow correction could be effective [98]. Gene therapy holds great promise in GD to reduce the need for expensive, lifelong treatments for patients with GD.

Another current aim of gene therapy for GD is to deliver a working gene copy to the CNS to alleviate nGD [16]. However, gene therapy to the CNS is an inherently difficult prospect, as it requires the use of BBB-penetrant vectors. Most gene therapy studies in clinical and preclinical settings use viral approaches. This review focuses on adeno-associated viruses (AAVs), which have shown the most promise in delivering genes to the CNS [97]. Additionally, there are many preclinical studies developing AAVs for GD due to their relatively low immunogenicity and highly specific tropism [98]. Recent advances in engineering of AAV capsid variants have led to the development of new serotypes that greatly increase viral transduction in the CNS [99,100], including serotypes 1, 2, 5, 8, 9 and 10 [100].

Delivery and transduction of AAV is highly dependent on the delivery mechanism used, which are highly varied and can be modified in conjugation with several technologies. Using intrathecal or intracerebroventricular (ICV) delivery, AAVs can be delivered directly into the CNS, physically bypassing the BBB [101]; thus, these injection routes are being considered for fetal gene therapy of GD2 [102]. However, these routes, especially ICV delivery (requiring surgical access to the cranial cavity), are highly invasive and may result in reluctance to undertake treatment [103]. A promising less invasive delivery route is to deliver BBB-penetrant AAVs like AAV9 intravenously [101]. Recent papers have used genetic engineering methods such as CREATE to modify AAV capsids to enhance transduction of various CNS cell types [104]. A recent study using a modified AAV9 targeting the human transferrin receptor resulted in successful IV delivery of *GBA1* into the CNS [99]. These recent results suggest that more invasive methods of delivery may be circumvented by targeted IV delivery in future clinical studies.

In AAV studies, the choice of promoter included in the transgene cassette is critical, as promoters help to determine the strength of expression of the gene, as well as the immune response from transduction [101]. Strong ubiquitous promoters such as cytomegalovirus (CMV), CAG, and CBA are common, as they drive high expression across a variety of cell types [105]. However, these promoters may cause off-target toxicity such as triggering hepatocellular carcinoma [106]. One strategy to avoid off-target toxicity is to use a cell-specific promoter. A recent study showed that an AAV using the Syn1 promoter, which targets neuronal cells, more effectively rescued the phenotype in a nGD mouse model than AAVs using CBA or CAG promoters, retaining higher brain GCase and lower activity in peripheral tissues [107]. EF1 promotors, while weaker, may be utilized to avoid toxicity caused by strong ubiquitous promoters. To enhance expression when weak promoters are selected, codon optimization [108], enhancer elements, or self-complementary AAV vectors can be considered [109].

Currently there are eight gene therapy clinical trials (see Table 1) using four AAV constructs for GD or *GBA1*-PD, but the serotypes, routes of delivery, and scope of treatment vary from study to study. Prevail Therapeutics is using an AAV9 vector encoding a *GBA1* transgene with both CMV and CBA genetic regulatory elements (PR001) targeting neurons [110]. Their PROVIDE trial (NCT04411654) administered PR001 in infants diagnosed with GD2 via intracisternal injection into the cisterna magna. Similarly, a clinical trial being performed in China (NCT06272149) employs an AAV9 vector, VGN-R08b, to treat infants with GD2 by ICV injection into the lateral ventricles. Additionally, this vector will be tested in a trial for GBA1-PD (NCT07414290). Two additional trials are evaluating PR001 administration in adult populations: the PROCEED trial (NCT05487599) investigates IV administration in a GD1 cohort, and the PROPEL trial (NCT04127578) utilizes intracisternal injection to access the CNS in patients with *GBA1*-PD. The ongoing GALILEO-2 (Phase 1/2, NCT06545136) and following GALILEO-3 (Phase

3, NCT07223944), utilize FLT201, an AAVS3 capsid with an engineered *GBA1* transgene targeting liver, which is delivered IV. This trial is for GD1, since the vector is unlikely to enter the CNS [111]. Another liver-based trial utilizes an AAV8 vector, LY-M001, intended for GD1 patients ages 6 to 18 years (NCT06528080) and ages 18 to 60 years (NCT06818838).

Looking ahead, the safety and efficacy from current clinical trials must be firmly established before the next generation of GD gene therapies can transition into clinical practice. One future consideration for GD2 is fetal gene therapy, since the accumulation of substrate begins early in gestation, and can result in non-viable pregnancies, hydrops fetalis, and neurological complications [4]. Preclinical studies of fetal gene therapy have been promising, including in non-human primates [102]. This strategy may prove useful in cases when the diagnosis of GD2 is established prenatally.

7. Gene editing

CRISPR/Cas9 editing technology has been applied to many genetic disorders, including GD, although the presence of the nearby homologous pseudogene renders editing in this locus challenging. However, CRISPR technology has been able to successfully correct some *GBA1* variants [112]. The potential use of ribonucleoproteins (RNP)/AAV6 platform and CRISPR/Cas9 to target GCase expression cassettes with monocyte/macrophage-specific elements to the CCR5 safe-harbor locus in human hematopoietic stem and progenitor cells has been successfully demonstrated *ex-vivo* [113]. Development of this homology-directed repair strategy allows for variant-independent correction.

Recently, using a new GD1 mouse with more progressive hematological and skeletal phenotypes [114], an RNP/AAV6 platform was used to deliver a functional copy of the human *GBA1* transgene into murine hematopoietic stem and progenitor cells (HSPCS). Performing autologous transplantation into GD1 mice successfully reversed the GD phenotype. This system was then applied to human HSPCs, potentiating a genome editing protocol for clinical application.

Allogenic cell transplantation of genome-edited cells reduces the risk of mutagenesis or immunogenicity and is a promising one-time alternative to current GD1 therapies, although it does not affect the neuronopathic manifestations of nGD [114].

8. Conclusion

The development of FDA-approved therapies for GD radically changed the therapeutic outlook for GD and other LSDs. However, aspects including expensive pricing, the inconvenience of chronic ERT infusions, lack of BBB penetrance, and the restricted eligibility criteria for SRT continue to pose challenges. Gene therapy and gene editing hold potential as a one-time treatment for GD, and clinical trials using gene therapy vectors are underway. However, the safety and tolerability of these therapies in patients must still be established. New research efforts focusing on identifying small molecule compounds that can cross the BBB have proven successful in many preclinical studies, but some have failed in early phases of clinical trials.

For rare diseases like GD and other LSDs, starting the drug development process from synthesis of novel compounds is expensive and slow. Large screens of FDA-approved compounds can identify structures that can be further tested in *in vitro* assays for GD. This approach allows for cost-effective and faster alternatives, as these pre-approved drugs have well characterized pharmacokinetics and safety profiles [115].

It is important to develop more predictive *in vivo* and *in vitro* preclinical methods to improve the likelihood of efficacy translating to clinical trials. Generating an accurate mouse model of GD and *GBA1*-PD has proven to be difficult, as models based on human mutations often do not recapitulate the human phenotype. Better models will be necessary for *in vivo* preclinical drug studies, particularly for investigating the mechanisms underlying *GBA1*-PD.

Most cell-based high-throughput screens use artificial substrates as activity read outs, which does not accurately reflect GCase activity in lysosomes. Endogenous GCase ceramides and newer live-cell substrates may be more valuable for validating drug candidates in future studies [71,116]. Most compounds tested in cell culture tend to be performed in 2D models, but 3D cell models are more physiologically relevant and can mimic the *in vivo* response, with potential to better predict drug efficacy [117]. 3D bioprinting is an emerging tool for high-throughput 3D models, which will be especially useful in future drug screening efforts.

Overall, while more research is needed to assess the long-term effects of these newly emerging therapies, the outlook for GD treatments is rich with promise.

9. Expert opinion

9.1. Progress

After 1991, with the approval of alglucerase as the first ERT for GD, the natural history of Gaucher disease underwent a dramatic change (Figure 2). The impacts of therapy have been life-altering for many patients with this disorder, with improvements in health, quality of life, and life expectancy. Currently, there are four different effective FDA-approved options for patients, a remarkable achievement. However, substantial unmet needs remain. The available treatments are extremely costly, some are inconvenient, and none are brain penetrant. Moreover, all are life-long therapies and not a 'cure.' Also, certain Gaucher complications, including pulmonary involvement, the development of gaucheromas (collections of Gaucher cells in tumor-like structures), protein-losing enteropathy, and certain bone lesions remain refractory to current therapies.

The association of *GBA1* with parkinsonism has revitalized research into new strategies to enhance GCase function. Currently, both academia and industry have active programs working to develop new therapeutics focusing on patients with *GBA1*-PD. On the horizon are likely to be novel schemes to deliver GCase to the brain to treat parkinsonism. Patients with neuronopathic GD, who have the greatest unmet needs, may be the first to benefit from the PD-focused efforts.

Multiple therapeutic concepts have already advanced from biological rationale to clinical implementation, including ERT, SRT, and some small molecules designed to improve GCase folding, trafficking, or function. Gene therapy is being developed for patients along the entire GD spectrum, including an early delivery strategy for affected fetuses. Clinical trials are underway and/or planned for babies with GD2, adolescents with GD3, both stable and treatment-naive patients with GD1 and even *GBA*1 heterozygotes with PD. These new approaches will certainly alter the disease landscape in the coming years.

9.2. Ethical considerations

Since the clinical outcomes in treated patients with GD1 have been so outstanding, there are ethical considerations regarding who should be recruited for trials of a new therapeutic. Both the risk-benefit ratio and the patient's personal outlook must be taken into consideration. It could be argued that new therapies should be first tested in patients inadequately controlled by current standard-of-care medications. But these patients tend to exhibit the most severe symptoms and many are infants or young children. Also, patients with the worst prognosis may not have reversible disease, which might lead to unsuccessful trials that would limit availability of drugs that could be beneficial for others.

9.3. Affordability

Multiple intervention points are available, ranging from improving residual enzyme performance to modulating proteostasis and lysosomal capacity, and restoring gene activity through gene transfer approaches. However, all currently approved medications targeting GD are extremely expensive, often costing hundreds of thousands of dollars per patient per year. In the future, affordability should be built into the design, not negotiated later through pricing and reimbursement. Repurposing provides a practical path to scalable therapies, particularly for CNS-relevant strategies that require repeated dosing and long durations. Ambroxol is a useful example of this approach. Early signals in neuronopathic GD support feasibility [118], and evidence of CNS exposure and target engagement in PD should justify further evaluation [79]. Randomized placebo-controlled trials are still needed to determine clinical efficacy and to define dosing and outcome measures both when considering this drug for GD or for PD. Other small molecules that serve as enzyme activators or chaperones should also be far less costly than drugs currently on the market. It could also be argued that while gene therapy is extremely expensive to develop and administer, if it succeeds as a 'once and done' treatment, it may still bring down the lifetime-associated costs.

9.4. Applicability to Parkinson disease

While there are likely to be some biological commonalities, it is premature to assume that strategies targeting *GBA1*-PD will necessarily be appropriate for GD. Currently the mechanism underlying *GBA1*-PD remains enigmatic. Also, there are challenges to trial design that are unique to PD. A central weakness is that trials often enroll heterogeneous patient groups, test drugs with different biological targets, and rely on endpoints that are not matched to the mechanism being tested. Target engagement is often treated as a surrogate for disease modification. In this context, target engagement refers to types of assays confirming enhanced whole-cell GCase activity or direct binding to GCase, complemented by functional assays assessing restoration of lipid homeostasis. However, especially for *GBA1*-PD, biochemical pathway modulation alone may not be sufficient to alter clinical trajectories, particularly when intervention occurs at symptomatic stages and endpoints are not tightly linked to the intended downstream biology. The MOVES-PD trial exemplifies this gap. Glucosylceramide synthase inhibition produced measurable substrate lowering but did not halt PD progression [52]. In parallel, uncertainty persists regarding the relative contribution of loss of enzymatic function, proteostatic stress from misfolded protein, and disrupted lysosomal capacity across phenotypes, which complicates selection of the optimal pharmacological profile. Drug development programs should therefore link pharmacology to disease-relevant downstream effects, align enrollment with disease stage, and select endpoints that can plausibly shift over the trial horizon.

The low penetrance of PD among *GBA1* carriers argues against indiscriminate preventive therapy. Biomarker-guided enrichment is needed to identify those individuals most likely to benefit, reduce overtreatment, and make trials feasible. The practical goal is to estimate the probability of future PD, conditional on what is already known about an individual, and then to enroll those with the highest predicted risk, rather than relying on a single biomarker threshold in isolation. When incidence is low, treating unselected populations requires large sample sizes and exposes many participants without benefit. Enrichment of high-risk groups markedly improves efficiency [119]. A comparable approach is needed for *GBA1*-PD, with biomarkers that classify state, stage, and rate, and that can also serve as treatment response readouts.

9.5. Tailored treatment rather than one size fits all

Ultimate goals differ by phenotype. For GD1, the main goal is sustained control of systemic multi-organ disease, including hematologic, visceral, and skeletal manifestations, while for the CNS manifestations of nGD, brain penetrance is the most critical aspect. Especially in the case of *GBA1*-PD, early disease modification is a priority, ideally before irreversible network or synaptic injury. Heterogeneity across *GBA1* variants also supports a shift toward mechanism-based matching. Small molecules that directly bind GCase, including activators and pharmacological chaperones, can selectively address specific defects, yet efficacy depends on the underlying biology. Some variants yield an enzyme that reaches the lysosome but has reduced catalytic function. Other variants are limited primarily by misfolding, ER retention, or accelerated degradation. Progress toward tailored therapy will require functional assays that classify variants by folding, trafficking, and activity under physiological conditions. Standardization is also needed to make results comparable across available compounds. Reporting should

include lysosomal readouts, intracellular target engagement, and correction of disease-relevant phenotypes beyond enzyme activity [69].

9.6. CNS delivery and cellular targets

BBB penetration for nGD and *GBA1*-associated PD is necessary but not sufficient. The dominant therapeutic target may differ by disease stage and phenotype. Neuron-centric strategies are supported by the linkage between GCase and α-synuclein biology, while glial and macrophage-lineage targets remain plausible given lipid handling defects and inflammatory signaling. Disease-relevant cellular systems should be prioritized in preclinical development. Patient-derived models that capture lipid accumulation phenotypes, such as iPSC-derived macrophages, and PD-relevant phenotypes, such as iPSC-derived dopaminergic neurons or microglia, are appropriate platforms for characterizing non-inhibitory chaperones and activators. Co-culture approaches may be needed to capture neuron-glia interactions that influence progression and treatment response.

9.7. Outlook

Over the coming years, an increased emphasis on BBB-penetrant small molecules is expected, with patient-relevant screening, biomarker-enriched clinical designs, and mechanistically aligned endpoints being critical features of their clinical testing. Combination strategies may be required when systemic and CNS correction must be achieved simultaneously. A translational pipeline that integrates functional variant classification, cell-type-specific testing, and biomarker selection for enrichment and treatment response readouts will likely determine which approaches will ultimately move from plausible to clinically meaningful.

The increased attention to *GBA1*-associated parkinsonism is already elucidating new roles for the protein and providing new insights into unanticipated pathways, disease modifiers and drug targets. This improved understanding of the cellular biology and pathophysiology associated with GCase deficiency will likely define which therapeutics will elicit meaningful improvements in patient outcomes and will be used for generations to come.

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